A Novel Approach to Detecting Muscle Fatigue Based on sEMG by Using Neural Architecture Search Framework

Shurun Wang, Hao Tang, *Member, IEEE*, Bin Wang, and Jia Mo

***Abstract*— Muscle fatigue detection is of great significance to human physiological activities, but many complex factors increase the difficulty of this task. In this article, we integrate several effective techniques to distinguish muscle states under fatigue and nonfatigue conditions via surface electromyography (sEMG) signals. First, we perform an isometric contraction experiment of biceps brachii to collect sEMG signals. Second, we propose a neural architecture search (NAS) framework based on reinforcement learning to autogenerate neural networks. Finally, we present an effective two-step training strategy to improve the performance by combining CNN with three types of commonly used statistical algorithms. Meanwhile, we propose a data enhancement algorithm based on empirical mode decomposition (EMD) to generate time-series data for expanding the dataset. The results show that this search algorithm can hunt for high-performing networks, and the accuracy of the best-selected model combined with support vector machine (SVM) for the group is 96.5%. With the same architecture, the average accuracy in individual models is 97.8%. The proposed data enhancement technique can effectively improve the fatigue detection performance, which allows further implementations in the human–exoskeleton interaction systems.**

***Index Terms*— Fatigue detection, human–exoskeleton interaction, neural architecture search (NAS) framework, surface electromyography (sEMG).**

I. Introduction

MUSCLE fatigue occurs after continuous exercise although people can maintain the activity in this state. In some industrial situations, workers have to carry weights repeatedly for a long time, so fatigue is a common phenomenon. However, researchers consider fatigue as the precursor of musculoskeletal diseases (MSDs) [1]. According to the working population reports of several surveys, the prevalence of upper limb symptoms of long-term manual workers is 20% ∼ 30% or even higher [2]. The assistance mode of the upper limb exoskeleton robot can reduce the risk of MSDs caused by repeated labor of workers, so it is urgent to establish a fatigue detection method for human–exoskeleton interaction [3].

Manuscript received 28 November 2019; revised 6 December 2020, 26 April 2021, and 8 October 2021; accepted 23 October 2021. Date of publication 9 November 2021; date of current version 4 August 2023. This work was supported by the National Key Research and Development Program of China under Grant 2017YFE0129700. *(Corresponding author: Hao Tang.)*

This work involved human subjects or animals in its research. Approval of all ethical and experimental procedures and protocols was granted by the Ethics Review Committee of the Hefei University of Technology.

The authors are with the School of Electrical Engineering and Automation, Hefei University of Technology, Hefei 230009, China (e-mail: wangsr@mail.hfut.edu.cn; htang@hfut.edu.cn; wang_bin@mail.hfut.edu.cn; jmo@mail.hfut.edu.cn).

Color versions of one or more figures in this article are available at https://doi.org/10.1109/TNNLS.2021.3124330.

Digital Object Identifier 10.1109/TNNLS.2021.3124330

So far, researchers have used a variety of methods to analyze muscle fatigue. Sonomyography (SMG) can describe the structural and morphological changes of skeletal muscle [4]. During exercise, muscle contraction causes changes in blood oxygen concentration and blood flow velocity. Noninvasive near-infrared spectroscopy (NIRS) techniques can detect fatigue by measuring the amount of oxygen in the muscle [5]. Mechanomyography (MMG) can record the vibration of muscle during muscle fiber movement and then observe the mechanical signals on the surface of the contracted muscle. It is also proved to be effective in measuring local muscle fatigue in different muscle groups [6], [7]. As a unique application of MMG, acoustic myography (AMG) can record the sound produced by muscle contraction, and some researchers compare it with surface electromyography (sEMG) [8]. The feature fusion of electroencephalography (EEG) and electrocardiography (ECG) for fatigue assessment has also achieved excellent results [9]. However, in various evaluation methods of human physiological information, sEMG has proven to be a mature and highly reliable noninvasive measurement method compared with other techniques [10]. sEMG can noninvasively measure the activation state of muscles, which is widely used in biomechanics and mechanical engineering, such as muscle fatigue detection, muscle force prediction, motion intention recognition, and exoskeleton robot control [11]–[14]. In view of these foundations, we focus on muscle fatigue detection based on continuously recording sEMG signals.

However, due to the weakness of sEMG signals, many factors affect the recording of its electrical signal activity, such as the duration and intensity of muscle contraction, electrode type, electrode placement position, external signal noise, and electrical properties of contact between electrode and skin [15]–[17]. Generally speaking, the electrical signal produced by skeletal muscle contraction is nonstationary and random, which can be sampled by using the surface electrode to get the muscle activity information [18]. Each person has different skeletal muscle characteristics, so it is not reliable to determine an accurate muscle fatigue threshold simply by the load and duration of muscle contraction. Therefore, in most cases, it is necessary to build the fatigue model of the subjects separately. Furthermore, muscle contractions include isometric contraction and dynamic contraction; different types of contractions can result in muscle fatigue [19]. Because of the sensitivity of sEMG, electrical activity signals produced by different actions are quite different, so it is necessary to determine a fixed scene for the fatigue detection of different muscle contractions.

2162-237X © 2021 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Despite these unfavorable factors, many studies have been conducted on the fatigue detection task. Traditional feature extraction methods usually start from the time domain, the frequency domain, the time–frequency domain, and the information domain [20]. In time-domain analysis, a combination of time-domain features is used for muscle pattern recognition [21]. The mean absolute value (MAV), the root mean square (rms) value, and the zero-crossing (ZC) rate are often used to assess muscle force [22]. In the frequency-domain analysis, the most common method is to obtain the median frequency (MDF) and mean power frequency (MPF) of the signal based on the power spectral density. These two features are considered to be effective in detecting fatigue [23], [24]. In the time–frequency-domain analysis, the Stockwell-transform (S-Transform), B-Distribution (BD), and extended modified BD can be used to detect muscle fatigue under dynamic contraction [25]. Wavelet transform is widely used to assess muscle fatigue and muscle force [26], [27]. In the information-domain analysis, using multiple time scales of information entropy, sample entropy, and fuzzy approximation entropy to analyze muscle fatigue shows certain advantages [28], [29]. At the same time, adopting the fusion of multiple features to assess muscle fatigue is used by many researchers. Feature fusion methods of the time domain and the frequency domain, such as rms and MDF, are used to detect the fatigue of triceps muscles [30]. In [31], 25 features are fused for fatigue classification. In general, designing an accurate fatigue detection model requires a solid foundation in signal analysis, which is a challenge for researchers.

In other studies, in order to distinguish nonfatigue and fatigue stages, hidden Markov model (HMM), genetic algorithm (GA), support vector machine (SVM), k-nearest neighbor (KNN), logistical regression (LR), random forest (RF), and other algorithms are reflected in recent studies [32], [33]. Different dimension reduction methods can be used to avoid overloading the classifier after the fusion of multiple features. Among them, the principal component analysis (PCA) and the independent component analysis (ICA) are widely used [34], [35]. At the same time, some researchers investigate the changing trend of different muscle characteristics in statistical significance, including the use of normalized MPF (NMPF), rms, and MDF [24], [30].

In recent years, due to the popularity of deep learning (DL), neural networks have also been used for physiological signal analysis [36], [37]. Some researchers adopt sEMG signals for fatigue state classification through DL architecture. In [38], a multilayer perception neural network (MLPNN) containing ICA was used for fatigue classification, but MLPNN was only used as a fitting tool after feature extraction, and the sEMG signal was not directly used as an input to the network. In [39], the deep belief network (DBN) was used to construct a fatigue model for the group population, but the DBN was too complex to adjust the parameters when designing the restricted Boltzmann manually. Different subjects are needed to adjust parameters separately when using DBN, so it is not appropriate to build a human fatigue model. Nevertheless, the convolutional neural network (CNN) can convert the original physiological signal into new expressions without significant information loss. Using CNN to detect driving fatigue based on EEG signals [40] and using MMG signals for intention recognition [41] have shown promising results. However, the network designer has to make numerous design choices to construct a proper CNN architecture, which is another challenge. In [42], neural architecture search (NAS) with reinforcement learning was proposed to generate networks for image classification, and the results were enlightening.

To the best of our knowledge, there is little literature on the application of NAS to assess upper limb muscle fatigue based on sEMG signals. Researchers with domain knowledge usually adopt handcrafted ways to extract features of physiological signals (e.g., EMG and MMG) in an explicit way, and it may take a long time to test and select suitable features for high classification performance, whereas CNN can learn to extract more discriminative features from the original signal in an implicit way. Unfortunately, the possible design space of CNN architectures is extremely large, and it is infeasible for an exhaustive manual search. Inspired by the NAS method in [42], we develop a hierarchical exploration mechanism for generating CNN architecture. Therefore, the method adopted in this article can be considered as the first attempt at generating a fatigue detection model for physiological signals. The novelties of this article are listed as follows.

1) We propose a hierarchical exploration mechanism based on reinforcement learning to automatically generate CNN networks, which can be used in the analysis of time-series signals.
2) An effective data enhancement algorithm based on empirical mode decomposition (EMD) is proposed to make up for the limited samples when building individual models.

The structure of this article is given as follows. First, we describe the implementation details of the improved NAS algorithm, the two-step training strategy, and the data enhancement method. Second, we present the performance of the proposed method compared with the traditional method and discuss it. Finally, we give conclusions with an outlook of future applications on human–exoskeleton interaction systems.

II. Methodology

In this section, we first present the basic designs and properties of the improved NAS framework. The two-step training strategy will be described in Section II-C. The data enhancement algorithm will be introduced in Section II-D.

A. Model Architecture Components

A typical CNN architecture consists of several convolution, pooling, and fully connected layers. In this work, we select the following components for constructing different CNN architectures.

1) Input Layer: The data input of sEMG signals can be represented by a pile of data segments as follows:

$$x_j^i = \left[o_j^i(1), o_j^i(2), \ldots, o_j^i(n)\right] \tag{1}$$

where $i = 1, 2, \ldots, D$, D is the signal dimension of sEMG signals, $j = 1, 2, \ldots, K$, K is the number of sample segments,

and n is the data position within each sample segment. In [43], sEMG signals are first transformed into a spectrum through the continuous Fourier transform and then extracted by convolution neural networks. In this article, sEMG signals are directly used as the input layer of networks, which can greatly reduce the complexity and execution time of the algorithm.

2) Convolutional Layer: Convolution kernels of different sizes in convolutional layers are responsible for learning-related features from the input data. The neurons of a feature map are only connected with the neighboring neurons of its previous layer. Such a neighboring area is called the local receptive field of the neuron in the previous layer. For a new feature map calculation, the process is the inner product of the local receptive field and the convolution kernel. Assume that, in CNN, n_{l-1} is used to represent the total number of feature maps of the $(l-1)$th layer, and the calculation formula of the convolutional layer is given as follows:

$$x_j^l = f(u) = f\left(BN_{\gamma,\beta}\left(\sum_{k=1}^{n_{l-1}} w_{jk}^l * x_k^{l-1} + b_j^l\right)\right) \tag{2}$$

where $w_{jk}^l \in R^2$ is the convolution kernel between the kth feature map of the $(l-1)$th layer and the jth feature map of the lth layer. b_j^l is the offset value of the jth feature map of the lth layer, x_j^l is the jth feature map of the lth layer, x_k^{l-1} is the kth feature map of the $(l-1)$th layer, $*$ is a convolution operation, $BN_{\gamma,\beta}(\cdot)$ is a batch normalization (BN) function, and $f(\cdot)$ is an activation function.

3) BN: After convolution and before activation function, BN can reduce the transfer of internal covariates and improve the generalization ability of the network. The formula of BN layer is given as follows:

$$u_i = \gamma \frac{x_i - \mu_b}{\sqrt{\sigma_b^2 + \varepsilon_{bn}}} + \beta. \tag{3}$$

The specific operation steps are to subtract the minibatch mean value from the feature map before inputting the activation function and then divide by the standard deviation. The data are limited to the mean value of 0 and the variance of 1, which reduces the expression ability of the network. To solve this problem, γ and β are used to scale and translate the overall data; they are learnable parameters. $B = \{x_1, x_2, \ldots, x_m\}$ is the input set of minibatch, μ_b and σ_b^2 are the mean and variance of the batch, and ε_{bn} is the constant term.

4) Pooling Layer: In order to further reduce the parameter number of the network connection, the pooling layer performs data compression on the convolved feature map to avoid overfitting of the model. This operation of reducing the feature resolution is called pooling as follows:

$$x_j^l = f\left(\lambda_j^l \mathrm{down}\left(x_j^{l-1}\right) + b_j^l\right) \tag{4}$$

where down$(\cdot)$ represents the sampling function under the pooling operation, which is generally divided into average pooling or maximum pooling. In order to make the model more learnable, a multiplicative factor λ_j^l and a bias b_j^l can be given for each pool feature map and then transferred to a nonlinear activation function $f(\cdot)$. Here, λ_j^l and b_j^l are the weight and offset of the jth feature map of the lth layer, respectively. Assuming that the pooled kernels of $m \times n$ perform nonoverlapping sliding on the input feature maps of $h \times w$, h and w can be divisible by m and n, respectively, the output feature maps' sizes are $(h/m) \times (w/n)$, and the scale invariance of the feature map can be maintained.

5) Fully Connected Layer: The number of input neurons is equal to the dimension of input data, and the number of output neurons in the classification task is equal to the number of output classes. The output of node j in layer l is

$$x_j^l = f\left(\sum_{k=1}^{n_{l-1}} w_{jk}^l x_k^{l-1} + b_j^l\right) \tag{5}$$

where w_{jk}^l is the connection weight between the kth node of the $(l-1)$th layer and the jth node of the lth layer, and b_j^l is the offset of the jth node of the lth layer.

6) Activation Function: The ReLU function is selected as the activation function of convolutional layer and dense layer. Compared with Sigmoid function, ReLU has the advantages of easy calculation and fast network convergence [44]. Its specific form is given as follows:

$$f(x) = \begin{cases} x, & x > 0 \\ 0, & x \le 0. \end{cases} \tag{6}$$

In Table I, we empirically list the parameter set associated with each layer type in the CNN and the allowed values for each parameter. By choosing different combinations, a very large number of networks can be generated.

B. Designing Network With Reinforcement Learning

In this article, the network is constructed by stacking different layers, so we represent it by a finite directed graph, and each layer can be considered as the graph node. We adopt a 6-D tuple to indicate the layer information of each graph node, the elements of the tuple are *Index*, *Type*, *Kernel Size*, *Kernel Number*, *Stride Size*, and *Neuron*, respectively. Among them, *Index* is set for recording the position of the current node. We define several ordered 6-D tuples as network structure codes; in this case, suitable structure codes will determine a neural network.

Although we have reduced the parameter space of the network structure, there are still a large number of potential networks to seek. Motivated by the MetaQNN proposed by Baker *et al.* [42], we propose a hierarchical exploration NAS framework and name it HEQNN. Our method that is based on Q-learning, a typical algorithm of reinforcement learning, consists of an *agent*, *states*, and *actions*. In a finite number of steps, the agent takes action to explore the environment based on its current state in order to obtain the maximum cumulative reward. In this article, we restrict the environment to obtain a discrete *state* space $\mathcal{S}$ and *action* space $\mathcal{A}$. The state $s \in \mathcal{S}$ represents the current graph node, and the *action* $a \in \mathcal{A}(s)$ is the decision for choosing the next successive graph node. Fig. 1(a) shows the state transition process $(s_t, a_t) \rightarrow (s_{t+1})$, where t is the current layer index. The state transition is stochastic, and it may be unknown to the agent. Fig. 1(b)

TABLE I
NETWORK STRUCTURE PARAMETER SPACE

Layer Name	Index	Type	Kernel Size	Kernel Number	Stride Size	Neuron
Input Layer (I)	$\{1\}$	1	0	0	0	0
Convolutional Layer (C)	$\{2, \cdots, N-1\}$	(2i, 2ii)	$\{6, 51, 101\}$	$\{6, 12, 18\}$	$\{1\}$	0
Pooling Layer (P)	$\{2, \cdots, N-1\}$	(3i, 3ii)	$\{2, 4, 8, 12, 16, 20\}$	0	As with Kernel Size	0
Fully Connected Layer (F)	$\{2, \cdots, N-1\}$	4	0	0	0	$\{10, 16, 32, 64, 128, 256\}$
Terminal (T)	$\{3, \cdots, N\}$	5	0	0	0	0

N is the max layer index.
(2i) means that BN layer is stacked after convolutional layer and (2ii) means that there is no BN.
(3i) means that there are learning parameters in pooling layer and (3ii) means that the pooling layer has no parameters.

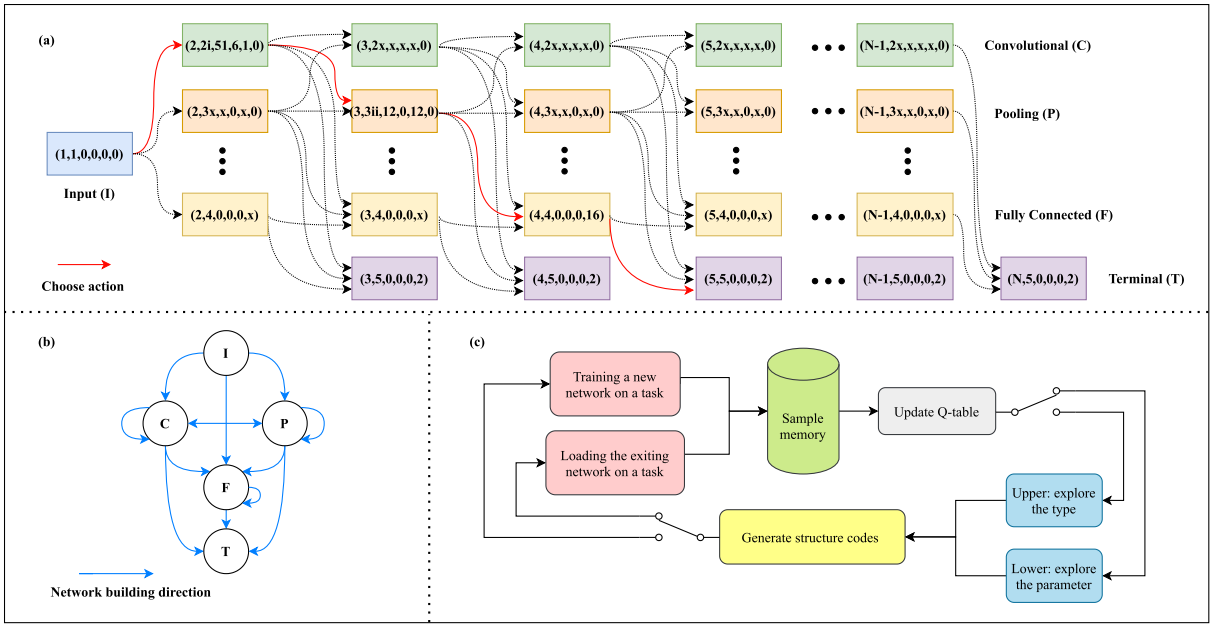

Fig. 1. Process illustration of NAS. (a) State transition process by different action choices. (b) Direction of the layer stack. (c) Flowchart of the HEQNN procedure.

shows the flow direction of the graph nodes; according to the combination of these graph nodes, a determined network structure can be obtained. The network structure can be considered as an action trajectory $\tau_{a_{1:n}}$, where n is the index of the terminal layer. To discover the best network structure, the agent maximizes its expected reward over all possible trajectories, expressed by R_τ

$$R_\tau = \mathbb{E}_{P(\tau_{a_{1:n}})}[\mathbb{R}] \tag{7}$$

where $P(\cdot)$ is the possibility and $\mathbb{R}$ is the cumulative reward. The recursive Bellman equation can be used to maximize the expected reward. At a time step t, given the current state s_t, the agent takes a subsequent action $a_t \in \mathcal{A}(s_t)$, arrives in next state s_{t+1}, and receives reward r_{t+1}. The maximum total expected reward can be denoted as $Q^*(s_t, a)$, and the Q value of this *state–action* pair combined with the Bellman equation can be written as

$$Q^*(s_t, a) = \mathbb{E}_{s_{t+1}|s_t,a}\Big[\mathbb{E}_{r|s_t,a,s_{t+1}}\big[r|s_t, a, s_{t+1}\big] + \gamma \max_{a' \in \mathcal{A}(s_{t+1})} Q^*\big(s_{t+1}, a'\big)\Big]. \tag{8}$$

Empirically, the solution of the Bellman equation can be formulated as an iterative update [45]

$$Q(s_t, a_t) = \big(1 - \alpha_q\big) Q(s_t, a_t) + \alpha_q\Big[r_{t+1} + \gamma_q \max_{a' \in \mathcal{A}(s_{t+1})} Q\big(s_{t+1}, a'\big)\Big] \tag{9}$$

where α_q is the learning rate, γ_q is the discount factor, and r_{t+1} is the immediate reward obtained from the environment by choosing *action* a_t in *state* s_t at the tth layer. Note that we use the validation accuracy of the candidate architecture as the reward. A predefined Q-table is commonly used to store all Q values.

Constructing a network architecture requires determining the layer type and the parameters of each layer. In [42], this required information is integrated together for exploration, which definitely increases the complexity of the searching process. Therefore, an effective mechanism that can enhance the exploration ability and reduce the futile exploration is essential. For the purpose of searching high-performing architectures, we propose a hierarchical exploration mechanism, which is combined with the Q-learning algorithm. We adopt an upper–lower relationship for exploration, with the agent performing an upper level exploration at the first phase, and performing a joint upper lower exploration after a certain stage. Both exploration strategies use the ε-greedy method [46]. During the upper level exploration, the agent explores the type of layers with the probability of ε_u and selects the best layer with the probability of $1 - \varepsilon_u$. In order to obtain more network architectures, the parameter set during each exploration is stochastic. The upper level exploration function is given as follows:

$$\varepsilon_u = \varepsilon_u^e + \big(\varepsilon_u^s - \varepsilon_u^e\big) \exp^{-t/\varepsilon_u^d}, \quad t \in (0, M_T) \tag{10}$$

Algorithm 1 Automatic Network Design With Q-Learning

Input: Initialize $Q \leftarrow (s, a), \forall s \in \mathcal{S}, \forall a \in \mathcal{A}(s) : 0.5$,
learning rate α_q, discount factor γ_q, total learning iteration M_T,
hierarchical exploration iteration M_1,
replay memory $\mathcal{P} \leftarrow \varnothing$
Output: Q.

for episode $= 1$ to M_T **do**
  obtain state sequence S and action sequence A by sampling a network according to the hierarchical exploration process (**Algorithm 2**)
  if the network is built successfully according to S **then**
    $accuracy \leftarrow$ Train(S) or Test(S)
  else
    $accuracy \leftarrow -1$
  end if
  if S is not be trained **then**
    $\mathcal{P} \leftarrow \mathcal{P} \cup (S, A, accuracy)$
  else
    replace the old accuracy with a new one
  end if
  for memory $= 1$ to K **do**
    $S_{sample}, A_{sample}, accuracy_{sample} \leftarrow$ Uniform($\mathcal{P}$)
    reward sequence $\leftarrow accuracy_{sample}$
    while not terminated **do**
      $Q(s_t, a_t) \leftarrow (1 - \alpha_q)Q(s_t, a_t) + \alpha_q[r_{t+1} + \gamma_q \max_{a'} Q(s_{t+1}, a')]$
    end while
  end for
end for

where ε_u^d is the decay factor of the upper level exploration process, ε_u^s is the start value, and ε_u^e is the end value. t is the current learning iteration, and M_T is the total iteration. From the M_1th iteration, the agent performs a joint upper–lower exploration. After the upper level exploration, the agent explores the parameter set of the layer with the probability of ε_l and chooses the best parameter set according to the Q-table with the probability of $1 - \varepsilon_l$. The lower level exploration function is given as follows:

$$\varepsilon_l = \varepsilon_l^e + \left(\varepsilon_l^s - \varepsilon_l^e\right) \exp^{-(t-M_1)/\varepsilon_l^d}, \quad t \in (M_1, M_T) \tag{11}$$

where ε_u^l is the decay factor of the lower level exploration process, ε_l^s is the start value, and ε_l^e is the end value. Compared with the exploration process of MetaQNN, this mechanism of hierarchical exploration is more relevant to the concept of neural network construction. The designing procedure of HEQNN is shown in Algorithm 1, and the concrete implementation of the hierarchical exploration process is shown in Algorithm 2.

Algorithm 2 Hierarchical Exploration Process

Input: Q, episode,
state sequence $S \leftarrow [s_{Input}]$, action sequence $A \leftarrow []$
Output: S, A

if episode $\in [1, M_1)$ **then**
  explore the type of the layer: update ε_u
  while $A[-1] \neq$ terminate **and** Length(A)$< N$ **do**
    $\delta_u \sim$ Uniform[0,1)
    if $\delta_u > \varepsilon_u$ **then**
      $a \leftarrow \arg\max_{a \in \mathcal{A}(S[-1])} Q[(S[-1], a)]$
      $s' \leftarrow$ Transition($S[-1], a$)
      $s' \leftarrow$ (the type of s', random parameter set under s')
    else
      $a \sim$ Uniform($\mathcal{A}(S[-1])$)
      $s' \leftarrow$ Transition($S[-1], a$)
    end if
    A.append(a)
    if $a \neq$ terminate **then**
      S.append(s')
    end if
  end while
end if
if episode $\in [M_1, M_T]$ **then**
  update ε_u
  explore the parameter set of the layer: update ε_l
  while $A[-1] \neq$ terminate **and** Length(A)$< N$ **do**
    $\delta_u, \delta_l \sim$ Uniform[0,1)
    if $\delta_u > \varepsilon_u$ **then**
      if $\delta_l > \varepsilon_l$ **then**
        $a \leftarrow \arg\max_{a \in \mathcal{A}(S[-1])} Q[(S[-1], a)]$
        $s' \leftarrow$ Transition($S[-1], a$)
      else
        $a \leftarrow \arg\max_{a \in \mathcal{A}(S[-1])} Q[(S[-1], a)]$
        $s' \leftarrow$ Transition($S[-1], a$)
        $s' \leftarrow$ (the type of s', random parameter set under s')
      end if
    else
      if $\delta_l > \varepsilon_l$ **then**
        $a \sim$ Uniform($\mathcal{A}(S[-1])$)
        $s' \leftarrow$ Transition($S[-1], a$)
        $s' \leftarrow$ (the type of s', best parameter set under s')
      else
        $a \sim$ Uniform($\mathcal{A}(S[-1])$)
        $s' \leftarrow$ Transition($S[-1], a$)
      end if
    end if
    A.append(a)
    if $a \neq$ terminate **then**
      S.append(s')
    end if
  end while
end if

C. Two-Step Training Strategy of CNN Combined With Statistical Algorithms

CNNs can effectively extract spatial information and share weights by using local connections, which significantly reduces the number of parameters [47]. However, when sEMG is applied to fatigue detection, the number of samples is always limited. In this case, CNN as a classifier has the overfitting problem, but the statistical algorithm is a good solution to solve the classification task [48]. As we know, CNN provides an effective feature extraction way by mapping high-dimensional data to low-dimensional space. Therefore, we use CNN to extract features for training the statistical algorithms.

Fig. 2 shows the core process of combining CNN with statistical learning methods, and the whole process is divided into two steps: training CNN and training statistical algorithms. Before the training, the sEMG signals after data preprocessing are randomly divided into training samples and testing samples. When the training of the CNN is completed, the network structure and parameters are both saved. Then, the CNN model is applied as a feature extractor, and the flattened 1-D feature vector is acted as the input for training the statistical algorithms. This two-step training strategy provides an effective combination between CNN and statistical algorithms.

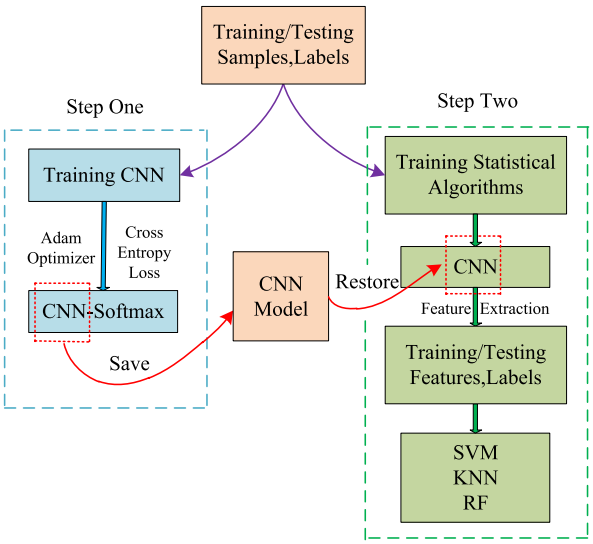

Fig. 2. Core process of the CNN model combined with statistical algorithms.

D. Data Enhancement

EMD is an adaptive signal analysis method proposed by Huang *et al.* [49]. This method can adaptively decompose a complex signal into residual components and a finite number of intrinsic mode functions (IMFs) whose frequencies are arranged from high to low according to the time scale characteristics of the signal. Each IMF contains the local characteristic of the original signal in different time scales. Meanwhile, the original signal can be recovered by adding all the IMFs and the residue without loss.

At present, researchers have adopted the EMD method to generate artificial EEG signals with good results [50], [51]. However, the traditional EMD method has the problem of modal aliasing in the decomposition process. The EEMD algorithm reduces the modal effect of EMD by adding noise [52], while the CEEMDAN algorithm proposed in [53] further reduces the modal effect by adding adaptive noise and has better convergence. Therefore, we use the CEEMDAN method to enhance the sEMG signal of the subject.

In Fig. 3, in the CEEMDAN decomposition procedure, we obtain several IMFs of the original signal by CEEMDAN. For our sEMG signal, we find that the decomposed signals have at most eight IMFs. Therefore, we take m as 8. Meanwhile, if the decomposed signals have less than eight IMFs, additional zero value IMFs will be added to reach eight IMFs for every decomposed signal. In the signal generation procedure, first, we define the number of artificial signals to ensure that each case (nonfatigue case and fatigue case) has the same number of artificial sEMG signals. Then, we randomly select m sEMG signals from the same case: the first selected signal contributes with its one random IMF; the second one contributes with its random IMF on different positions and successively until the mth signal. Finally, we generate a new one-channel sEMG signal by adding up all the IMFs.

E. Training Details

1) Experience Replay: Same as [42], we employ a replay memory to store the network structure and its corresponding validation accuracy after each iteration. All network structures in the replay memory do not duplicate each other, and the agent randomly samples 64 models from the memory and updates its Q values at a given interval. The flowchart of the HEQNN learning procedure is shown in Fig. 1(c).

2) Parameter Setting: During the architecture searching phase, each model is trained for a total of 8 epochs with the Adam optimizer [54] with a fixed learning rate of 0.001. The cross-entropy objective function is used as the loss function, the weights are initialized according to the Xaiver method [55], and the offset values are set to 0. The batch size is set to 64. To balance the relationship between exploration and exploitation, we decrease ε_u and ε_l following an exponential function. ε_u^e and ε_l^e are set to 0.05, and ε_u^s and ε_l^s are set to 0.95. When updating ε_u and ε_l, ε_u^d and ε_l^d take 100 and 300, respectively. The max layer index is set to 10, and the maximum number of visits to any layer of C, P, and F does not exceed three. Moreover, the total learning iteration is set to 2000, and the start exploration iteration of the lower level is set to 400. We choose α_q as 0.01 and γ_q as 1.0 [42]. For statistical algorithms, the kernel function of SVM is the radial basis function, the number of KNNs is 10, and the maximum depth of RF is 100.

3) Other Details: When generating the structure code, the Relu function is stacked after the convolutional layer. The pooling layer uses a nonoverlapping sliding pooling kernel and does not fill the edge with an average pooling operation of 0. In addition, the dropout technique is used after the fully connected layer to reduce the risk of overfitting [56], and half of the neurons are randomly activated per training session.

During the searching phase, the dataset without any data enhancement is divided into training and test set with the ratio of 4:1. In each model training stage, 1/3 random test data are selected to validate the performance. Within the given iterations, the replay memory is maintained to store the entire network (topology & parameters) and the prediction performance. Particularly, if a model is resampled that has already been trained, the saved parameters will be loaded and revalidate the model performance with 1/3 random test data. If the result is worse than before, we reduce the learning rate by a factor of 0.8 and restart training, with a maximum of three times for retraining operation.

After the searching phase, the top ten models are founded for fine training. We reduce the learning rate by a factor of 0.8 every 5 epochs with a maximum of 30 epochs. Finally, the top five models are saved for further validation. Our algorithm works on Intel CPU (Xeon Silver 4214R, 2.4 GHz) and NVIDIA GPU (Tesla A10) workstations using pytorch[1] and sklearn[2] libraries.

III. Experimental Setup

In order to investigate the characterization of upper limb muscle fatigue, we carefully performed an isometric contraction experiment of the biceps brachii. The valuable original dataset is generated by the sEMG signal. In our experiment,

[1]https://pytorch.org/
[2]https://scikit-learn.org/

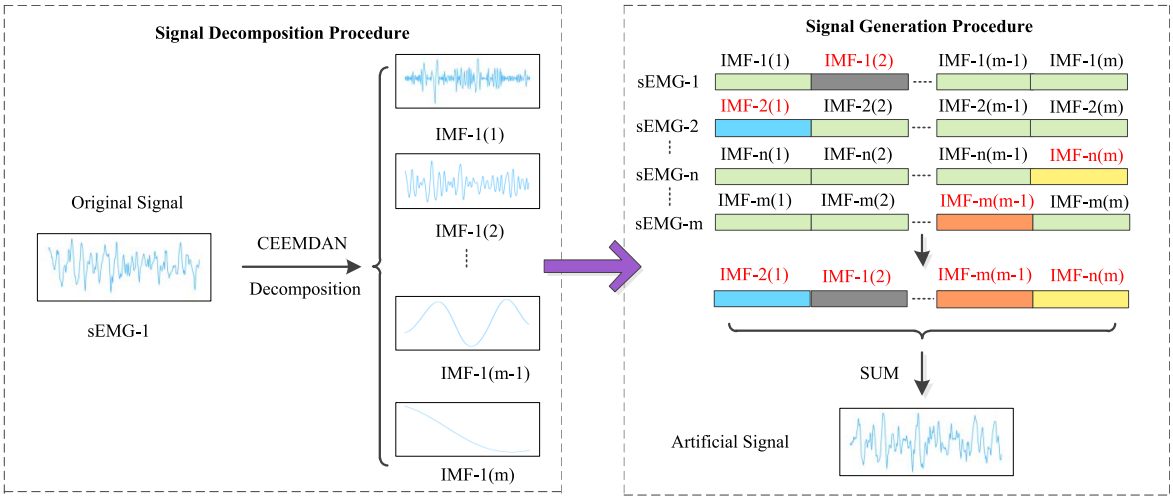

Fig. 3. EMD procedure of an original sEMG signal and the generation procedure of an artificial signal.

we fully considered the external factors of the experiment (e.g., indoor temperature and surrounding sound noise). In this section, we will introduce the subject information, data collection, and fatigue classification, respectively.

A. Participants

We recruited 12 healthy male volunteers who had no upper extremity bone disease. All subjects are right-handed with an average age of 23.5 ± 1.8 years for the purpose of reducing the influence of individual differences. One day before the experiment, all subjects were informed to have enough rest. All subjects were given written consent to participate in the experiment, and the content was approved by the Ethics Review Committee of the Hefei University of Technology.

B. Data Acquisition and Preprocessing

Before the experiment, a quiet and closed room at a warm temperature was arranged, and the biceps brachii was shaved and wiped with alcohol. A disposable Ag/AgCl differential electrode (5 mm in diameter) was attached to the muscle abdomen, and the distance between the electrodes was 2 cm [57]. The reference electrode and the angle sensor were placed at the proximal end of the elbow. To record the one-channel sEMG signal under isometric contraction of biceps brachii, we required the subject to stand vertically on a wooden platform with the arms parallel to the body, holding a 4-kg dumbbell and keeping the arm at 90° to the forearm. The digital display interface of an angle sensor is placed in front of the subject, which is necessary for the subject to maintain the posture until exhaustion.

Using the Ultium EMG sEMG system produced by NORAXON Company for data acquisition, the wireless surface EMG sensor can accurately measure and monitor the sEMG signal. The internal resolution of the system is 24 bits, the baseline noise is less than 1 μV, and the common-mode rejection ratio is not less than 100 dB. The system sampling frequency is set to 2000 Hz, and the gain used in this experiment is 1000. In order to remove movement artifact and 50-Hz power frequency interference, the original signal needs

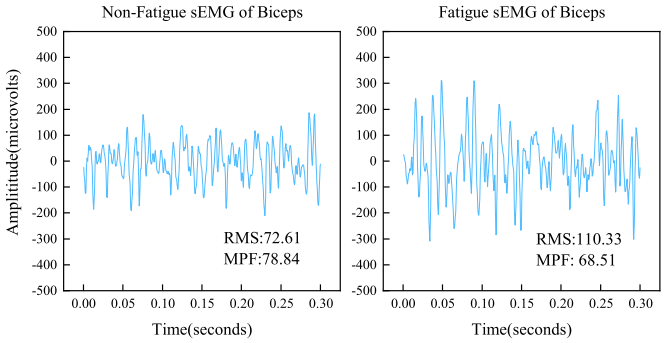

Fig. 4. Typical raw electromyographic signal of biceps brachii under isometric contraction.

to go through 20–400-Hz bandpass filtering and 50-Hz notch filtering [58]. The whole sEMG signal is processed by de-ECG and max–min normalization.

C. Fatigue Classification

The processed sEMG signal is divided into nonfatigue and fatigue cases [59] to train the fatigue classifier. According to the experimental protocol of most fatigue detection researchers [18], [25], [39] and the subjective feelings of the subjects [60], we define the initial stage of an experiment as the muscle nonfatigue stage, in which subjects can easily stabilize the weight. At the end of each experiment, the subjects struggled to resist the pressure of the weight, so this period was marked as the fatigue stage. The first 0.5 s and last 0.5 s of the sEMG signal were removed to reduce the action interference of the subjects at the beginning and end of the experiment. We take a 12-s dataset in each stage, the sample segment is 0.3 s, the sliding window is 0.1 s, each sample segment has 600 data points, and each stage has 118 sample segments. The typical raw signals of nonfatigue and fatigue stages are shown in Fig. 4. It shows that the data frames in the nonfatigue stage and fatigue stage are nonstationary, and the signal changes are more intense in the fatigue stage. The reason is that the subject has to increase the muscle force to resist the object's weight to maintain the posture in the fatigue stage. By evaluating the rms value and the average power frequency (MPF), it can be seen that the amplitude of the sEMG increases, and the frequency

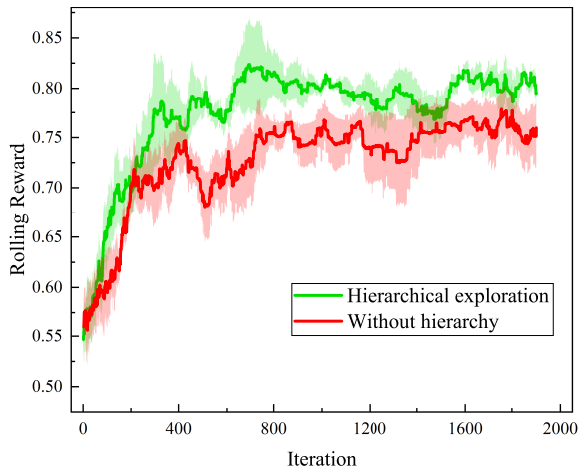

Fig. 5. Learning process of NAS. The lines show the average of five times learning process, and the shadow is the standard deviation.

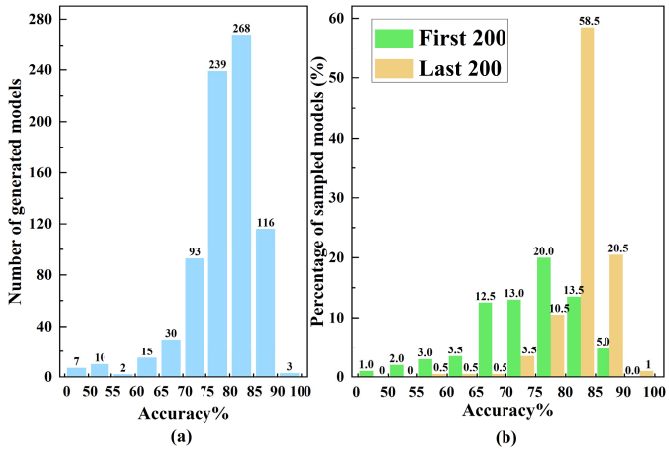

Fig. 6. Model distribution versus accuracy. (a) Number of generated models versus accuracy. (b) Percentage of sampled models versus accuracy.

spectrum decreases with the deepening of muscle fatigue. The mean endurance time is 126.42 s, and the standard deviation is 37.89 s.

IV. Results and Discussion

In this section, we first obtain the top five models by using the HEQNN algorithm for the whole subject group. Then, the best model is selected to detect the fatigue for each subject, and the data enhancement method is used to improve the detection performance. Finally, we analyze the advantages of our proposed algorithms and give a detailed discussion.

A. Network Searching Analysis

Fig. 5 compares the learning process of the two exploration mechanisms, it shows a rolling mean of the reward versus iteration, and the rolling window is 100. The plots present that the rolling reward goes up as ε decreases, demonstrating that the agent learns to choose better performing CNN architectures. In addition, hierarchical exploration reveals better convergence and a certain stability in the training process. The method without hierarchy tends to fall into local optima. Fig. 6 presents the model distribution versus accuracy in a complete training session, 783 models were successfully built, and the performance was evaluated. Fig. 6(a) shows that about half of the models can achieve more than 80% accuracy, while the accuracy of the rest models is mainly distributed in the range of 60%–80%. Fig. 6(b) provides the evidence that the agent is capable of selecting better performing architectures after sufficient exploration. In the first 200 iterations, 26.5% of the models cannot be successfully constructed. In the last 200 iterations, about 80% of the models get an accuracy of higher than 80%, and only 4.5% of the models are meaningless.

Table II provides the top five model architectures selected by NAS and the designed model by us. This table is quite revealing in several ways. First, there is no such architecture consisting of only fully connected layers. Second, Model4 reveals that the fully convolutional network can also achieve a good result. Third, multiple convolutional layers followed by pooling and fully connected layers perform better in fatigue detection tasks, and adding learning parameters when building pooling layers, indeed, improves the accuracy. Finally, the self-designed network also has certain competitive advantages.

B. Fatigue Detection Results for the Group

In addition to accuracy, AUC and F1 score are commonly used to analyze the results of the classification tasks. AUC is a kind of performance indicator to measure the advantages and disadvantages of the binary-classification model. The larger the value is, the better the model will be. F1 score is based on the harmonic average of precision and recall. Fig. 7 compares the experimental data on three types of performance indicators. Fig. 7(a) shows that most models combined with SVM (CNN-SVM) perform well; among them, Model1 combined with SVM achieves the highest accuracy of 0.965. Moreover, it can be found that the CNN-KNN performs not well under Model4, indicating that the network structure may influence the combination effect. From Fig. 7(b) and (c), there is no doubt that CNN-SVM is the most stable combined architecture; the highest values of AUC and F1 score under Model1 are 0.9906 and 0.9633, respectively. Fig. 7(d) provides the parameter numbers of each model; it can be seen that the fully convolutional network has more parameters than other networks, as there is no fully connected layer to reduce the dimension before output. It is worth noting that the designed network has the lowest number of parameters and good detection performance, indicating that it is also commendable to design a network empirically.

C. Fatigue Detection Results for the Individual

It can be seen from the previous introduction that the traditional feature extraction methods of sEMG signal usually extract statistical features from the time domain, the frequency domain, the time–frequency domain, or the information domain as the classification basis and can achieve good classification results. However, due to the complexity of nonstationary random signals, it is impossible to determine an optimal classification model by a specific feature or a combination of several features. For comparison, a total of 16 commonly used traditional features are selected, and the reconstructed feature vector is dimensionally reduced by PCA.

TABLE II
MODEL ARCHITECTURES

Model	Architecture	Accuracy
Model1	[I(1,1,0,0,0,0), C(2,2ii,101,18,1,0), C(3,2i,51,12,1,0), C(4,2i,51,12,1,0), P(5,3i,12,0,12,0), F(6,4,0,0,0,32), T(7,5,0,0,0,2)]	0.9167
Model2	[I(1,1,0,0,0,0), C(2,2ii,51,18,1,0), C(3,2i,51,6,1,0), C(4,2i,51,6,1,0), P(5,3i,16,0,16,0), F(6,4,0,0,0,64), T(7,5,0,0,0,2)]	0.9068
Model3	[I(1,1,0,0,0,0), C(2,2ii,51,18,1,0), C(3,2i,51,18,1,0), C(4,2ii,51,6,1,0), P(5,3i,16,0,16,0), T(7,5,0,0,0,2)]	0.9054
Model4	[I(1,1,0,0,0,0), C(2,2ii,51,18,1,0), C(3,2i,101,18,1,0), C(4,2i,51,12,1,0), T(7,5,0,0,0,2)]	0.9011
Model5	[I(1,1,0,0,0,0), C(2,2ii,101,18,1,0), C(3,2ii,101,6,1,0), C(4,2i,51,6,1,0), P(5,3i,16,0,16,0), F(6,4,0,0,0,16), T(7,5,0,0,0,2)]	0.8785
Designed	[I(1,1,0,0,0,0), C(2,2i,101,12,1,0), P(3,3i,20,0,20,0), C(4,2i,6,6,1,0), P(5,3i,2,0,2,0), F(6,4,0,0,0,10), T(7,5,0,0,0,2)]	0.8928

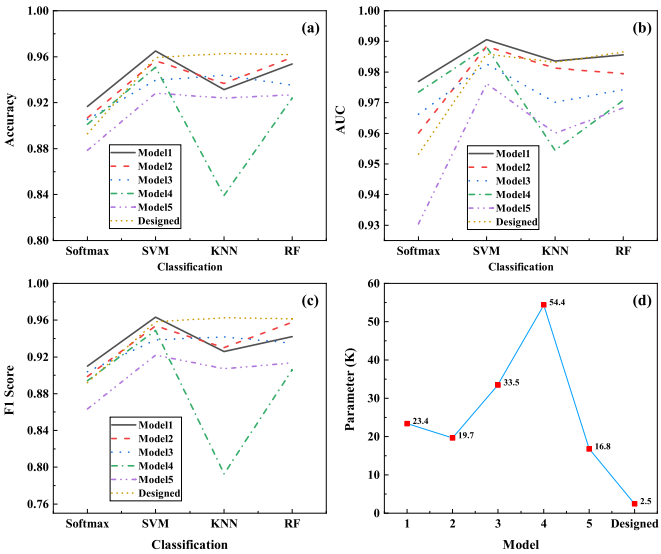

Fig. 7. Detection performance of the models combined with statistical algorithms. (a) Accuracy. (b) AUC. (c) F1 score. (d) Parameter number.

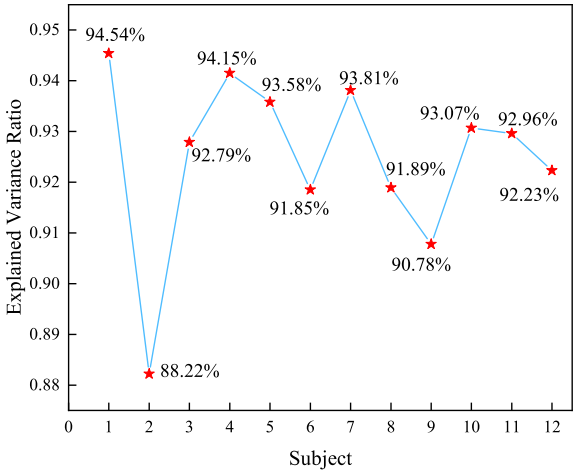

Fig. 8. Principal component variance contribution rate after PCA.

These 16 features are rms value, MAV, ZC rate, waveform length, mean square error, slope sign change number, variance, MPF, MDF, spectrum mean value, spectrum variance, spectrum entropy, spectrum energy, spectrum skewness, spectrum kurtosis, and marginal spectrum entropy. The fatigue perception of the subjects in the experiment process is different, and the corresponding results of feature extraction are also different. Considering the algorithms' training speed and dimensionality reduction effect, we select the first five main components after PCA. Furthermore, the PCA contribution rate represents the proportion of variance cumulative value of principal components in the overall components. Fig. 8 shows the line chart of the PCA variance contribution rate. The larger the value is, the more representative the principal components are.

Fig. 9 shows the performance of Model1 for the individual. Combining Figs. 8 and 9 to analyze and compare the fatigue detection accuracy, we find that, among all statistical algorithms after traditional feature extraction, the accuracy of PCA-SVM is in the leading position, followed by PCA-KNN. In the traditional feature extraction of subject 2, the contribution rate of PCV accumulated by the first five principal components is less than 90%, and the loss of some information may lead to low classification accuracy. Compared with other subjects, when the contribution rate of PCV is higher than 90%, the Pearson coefficient of the contribution rate of PCV and the accuracy of PCA-SVM is -0.0465. It means that there is no consistency between the accuracy of the fatigue model of different subjects and the contribution rate of PCV after dimensionality reduction. This is because when the principal component accounts for a large proportion of the total component, the accuracy of the individual fatigue model is mainly affected by individual differences. Moreover, the accuracy values of ten subjects are above 90% by using PVC-SVM: one is only 70%, and the others are 80%.

It is worth noting that CNN-Softmax does not perform well in some subjects, which may be due to the limited dataset. However, when CNN architecture is used as the feature extractor and the statistical algorithm as the classifier, the accuracy values of all subjects are improved. Surprisingly, the detection results of subjects 2 and 3 are greatly increased, which shows that CNN combined with statistical algorithms can make up for the shortcomings of traditional feature engineering and achieve excellent detection results. Moreover, it is found that the average accuracy of the individual under CNN-SVM is 97.8%, which is higher than the group. The reason is that establishing the individual fatigue model does not need to consider the difference of physiological information of different subjects, so it is more accurate than the group model.

To further verify the benefits of CNN combined with statistical algorithms in limited samples, we applied the CEEMDAN method for expanding the original dataset to train the network. Therefore, the sEMG signals are chosen from subjects 2 and 3, respectively. The mean and standard

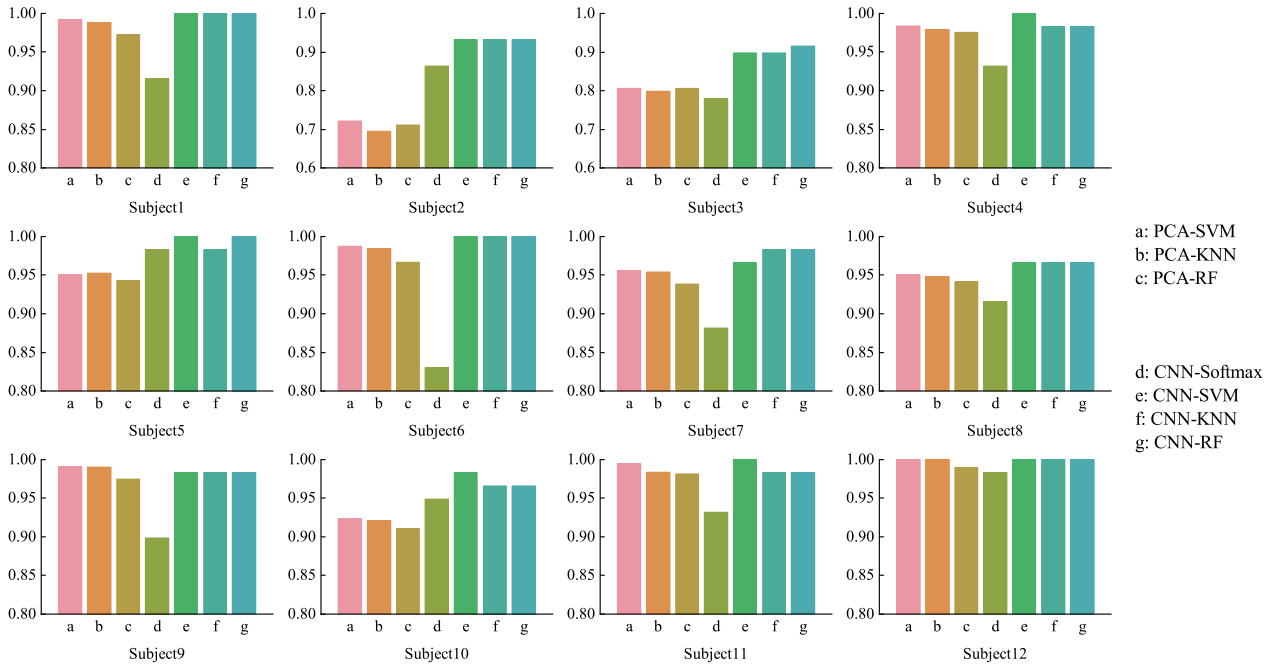

Fig. 9. Accuracy of the individual fatigue model.

TABLE III
ACCURACY OF FATIGUE DETECTION UNDER DATA ENHANCEMENT

Subject		Original Dataset				+1*AD	+2*AD	+3*AD
		a	b	c	d	a	a	a
2	mean	0.86	0.93	0.93	0.93	0.92	0.93	0.95
	std	0.06	0.02	0.03	0.03	0.03	0.02	0.01
3	mean	0.78	0.89	0.89	0.91	0.90	0.91	0.92
	std	0.06	0.04	0.04	0.05	0.02	0.02	0.02

AD: Artificial dataset with same numbers of the original dataset
a: CNN-Softmax b: CNN-SVM c: CNN-KNN d: CNN-RF

deviation of tenfold cross validation are given in Table III. It is obvious that data enhancement can improve the accuracy of the CNN model effectively, and the standard deviation is smaller. Among them, the larger the dataset, the higher the accuracy.

D. Discussion

In HEQNN, the agent learns to construct the network via hierarchical exploration and exploitation; it is natural to ask whether the agent can select the best network. In fact, despite limiting the number of optional parameters in the network setup and minimizing the complexity of the construction process, there will still be a vast number of model combinations. Among them, the best models may not be explored, but the agent will move toward the proper direction for network construction by using the proposed approach, and the final result may differ from the human-designed model. In addition, the human-designed model, although sometimes satisfying the task requirements, also takes some time for structural design and parameter optimization.

The feature extraction results of the sEMG signal are affected by the gender, health, and age of the subject [39]. To reduce the influence of individual differences, the physiological characteristics of all subjects should be similar when establishing group fatigue models [38]. Therefore, we recruited the subjects from our lab for the experiments, which are all master students of a similar age. The sEMG signal is consistent when used to detect individual fatigue, so our method is not affected by people of all ages and muscles in all places when establishing individual fatigue models. It remains to be studied to establish a model suitable for different age groups at the same time.

According to the comparative analysis of the experimental results, CNN has obvious advantages in fatigue detection compared with the traditional methods. Through the method of combining statistical algorithms, the classification performance is further improved. It shows that this combination architecture is effective for small sample classification. In addition, adopting the data enhancement method can also effectively improve fatigue detection accuracy. The increased amount of dataset allows CNN to learn the spatial and temporal information of sEMG signals more effectively.

Due to the data-driven characteristics of the model, the architecture maintains enough flexibility to deal with the detection task of muscle fatigue. In the results, we demonstrate that the introduction of adding parameters in the pooling layer is helpful to improve the network performance. Commonly in DL, the use of the pooling layer can greatly reduce the number of parameters, so the traditional CNN pooling layer does not need learning parameters. However, in our architecture, the network is concise in structure, so adding learning parameters in the pooling layer will not cause parameter redundancy but improve the performance of the model. Compared with traditional methods, the architecture search method does not need to rely on complicated signal processing and expert experience, which saves much feature-engineering time.

V. CONCLUSION

Muscle fatigue of workers in industrial production is a social problem that needs more attention. Traditional feature

extraction methods inevitably need to design features manually, which often leads to inadequate feature extraction and low classification accuracy. We propose a hierarchical exploration mechanism based on reinforcement learning to automatically construct a proper network for detecting fatigue, and the autogenerated networks can achieve more satisfactory results compared to self-designed networks. Meanwhile, the two-step training strategy of CNN combined with statistical algorithms and the data enhancement method both contribute to improve the detection performance.

The proposed methods have great application potentials, such as intention recognition and fatigue detection in human–exoskeleton interaction. Moreover, our methods are not limited to sEMG, and other time-series signals can also be used as the observation for fatigue detection. Further work will focus on multisource information fusion, hoping to further improve the performance of the model. Moreover, we will verify the applicability of the algorithm to other muscles. Looking forward to the future, we hope that the model selected by NAS can be used in dynamic real-time human–exoskeleton machine systems effectively.

Acknowledgment

The authors are also grateful to all the student population who participated in the study.

References

[1] S. Boyas and A. Guével, "Neuromuscular fatigue in healthy muscle: Underlying factors and adaptation mechanisms," *Ann. Phys. Rehabil. Med.*, vol. 54, no. 2, pp. 88–108, Mar. 2011.

[2] L. Punnett and D. H. Wegman, "Work-related musculoskeletal disorders: The epidemiologic evidence and the debate," *J. Electromyogr. Kinesiol.*, vol. 14, no. 1, pp. 13–23, Feb. 2004.

[3] Y. Roquelaure *et al.*, "Epidemiologic surveillance of upper-extremity musculoskeletal disorders in the working population," *Arthritis Rheumatism*, vol. 55, no. 5, pp. 765–778, Oct. 2006.

[4] J. Shi, Q. Chang, and Y.-P. Zheng, "Feasibility of controlling prosthetic hand using sonomyography signal in real time: Preliminary study," *J. Rehabil. Res. Dev.*, vol. 47, no. 2, pp. 87–98, 2010.

[5] Y. Yoshitake, H. Ue, M. Miyazaki, and T. Moritani, "Assessment of lower-back muscle fatigue using electromyography, mechanomyography, and near-infrared spectroscopy," *Eur. J. Appl. Physiol.*, vol. 84, no. 3, pp. 174–179, Mar. 2001.

[6] T. Kimura, M. Fujibayashi, S. Tanaka, and T. Moritani, "Mechanomyographic responses in quadriceps muscles during fatigue by continuous cycle exercise," *Eur. J. Appl. Physiol.*, vol. 104, no. 4, pp. 651–656, Nov. 2008.

[7] A. Jaskólska, P. Madeleine, A. Jaskólski, K. Kisiel-Sajewicz, and L. Arendt-Nielsen, "A comparison between mechanomyographic condenser microphone and accelerometer measurements during submaximal isometric, concentric and eccentric contractions," *J. Electromyogr. Kinesiol.*, vol. 17, no. 3, pp. 336–347, Jun. 2007.

[8] A. A. Rodriquez, J. C. Agre, E. R. Knudtson, Cota, T. M. Franke, and A. V. Ng, "Acoustic myography compared to electromyography during isometric fatigue and recovery," *Muscle Nerve*, vol. 16, no. 2, pp. 188–192, Feb. 1993.

[9] P. Xie, M. S. Qi, Y. Y. Zhang, Z. J. Liu, and S. C. Chen, "Driver fatigue assessment based on multi-physiological signals and transfer learning," *Chin. J. Sci. Inst.*, vol. 39, no. 10, pp. 223–231, Oct. 2018.

[10] M. R. Al-Mulla, F. Sepulveda, and M. Colley, "A review of non-invasive techniques to detect and predict localised muscle fatigue," *Sensors*, vol. 11, no. 4, pp. 3545–3594, Mar. 2011.

[11] J. Murillo-Escobar *et al.*, "Muscle fatigue analysis during dynamic contractions based on biomechanical features and permutation entropy," *Math. Biosci. Eng.*, vol. 17, no. 3, pp. 2592–2615, 2020.

[12] R. Ma, L. Zhang, G. Li, D. Jiang, S. Xu, and D. Chen, "Grasping force prediction based on sEMG signals," *Alexandria Eng. J.*, vol. 59, no. 3, pp. 1135–1147, Jun. 2020.

[13] X. Li *et al.*, "Towards reducing the impacts of unwanted movements on identification of motion intentions," *J. Electromyogr. Kinesiol.*, vol. 28, pp. 90–98, Jun. 2016.

[14] Z. Li, B. Wang, F. Sun, C. Yang, Q. Xie, and W. Zhang, "SEMG-based joint force control for an upper-limb power-assist exoskeleton robot," *IEEE J. Biomed. Health Informat.*, vol. 18, no. 3, pp. 1043–1050, May 2014.

[15] M. B. I. Reaz, M. S. Hussain, and F. Mohd-Yasin, "Techniques of EMG signal analysis: Detection, processing, classification and applications," *Biol. Procedures*, vol. 8, no. 1, pp. 11–35, Dec. 2006.

[16] D. J. Hewson, J.-Y. Hogrel, Y. Langeron, and J. Duchêne, "Evolution in impedance at the electrode-skin interface of two types of surface EMG electrodes during long-term recordings," *J. Electromyogr. Kinesiol.*, vol. 13, no. 3, pp. 273–279, Jun. 2003.

[17] X. Zhang *et al.*, "Research on gesture definition and electrode placement in pattern recognition of hand gesture action SEMG," in *Proc. Int. Conf. Med. Biometrics*, Berlin, Germany, Jun. 2008, pp. 33–40.

[18] G. Venugopal, M. Navaneethakrishna, and S. Ramakrishnan, "Extraction and analysis of multiple time window features associated with muscle fatigue conditions using sEMG signals," *Expert Syst. Appl.*, vol. 41, no. 6, pp. 2652–2659, May 2014.

[19] A. Georgakis, L. K. Stergioulas, and G. Giakas, "Fatigue analysis of the surface EMG signal in isometric constant force contractions using the averaged instantaneous frequency," *IEEE Trans. Biomed. Eng.*, vol. 50, no. 2, pp. 262–265, Feb. 2003.

[20] M. González-Izal, A. Malanda, E. Gorostiaga, and M. Izquierdo, "Electromyographic models to assess muscle fatigue," *J. Electromyogr. Kinesiol.*, vol. 22, no. 4, pp. 501–512, Aug. 2012.

[21] D. Tkach, H. Huang, and T. A. Kuiken, "Study of stability of time-domain features for electromyographic pattern recognition," *J. Neuroeng. Rehabil.*, vol. 7, no. 1, p. 21, May 2010.

[22] A. Phinyomark, F. Quaine, S. Charbonnier, C. Serviere, F. Tarpin-Bernard, and Y. Laurillau, "Feature extraction of the first difference of EMG time series for EMG pattern recognition," *Comput. Methods Program. Biomed.*, vol. 117, no. 2, pp. 247–256, Nov. 2014.

[23] S. Thongpanja, A. Phinyomark, P. Phukpattaranont, and C. Limsakul, "Mean and median frequency of EMG signal to determine muscle force based on time-dependent power spectrum," *Electron. Electr. Eng.*, vol. 19, no. 3, pp. 51–56, Mar. 2013.

[24] I. Ahmad and J.-Y. Kim, "Assessment of whole body and local muscle fatigue using electromyography and a perceived exertion scale for squat lifting," *Int. J. Environ. Res. Public Health*, vol. 15, no. 4, p. 784, Apr. 2018.

[25] P. A. Karthick, D. M. Ghosh, and S. Ramakrishnan, "Surface electromyography based muscle fatigue detection using high-resolution time-frequency methods and machine learning algorithms," *Comput. Methods Programs Biomed.*, vol. 154, pp. 45–56, Feb. 2018.

[26] L. M. Hari, G. Venugopal, and S. Ramakrishnan, "Muscle fatigue analysis in biceps brachii surface electromyography signals using synchrosqueezed Morlet wavelet and singular value decomposition," *Electron. Lett.*, vol. 57, no. 1, pp. 42–44, Jan. 2021.

[27] S. K. Chowdhury, A. D. Nimbarte, M. Jaridi, and R. C. Creese, "Discrete wavelet transform analysis of surface electromyography for the fatigue assessment of neck and shoulder muscles," *J. Electromyogr. Kinesiol.*, vol. 23, no. 5, pp. 995–1003, Oct. 2013.

[28] S. Wang, H. Tang, B. Wang, and J. Mo, "Analysis of fatigue in the biceps brachii by using rapid refined composite multiscale sample entropy," *Biomed. Signal Process. Control*, vol. 67, May 2021, Art. no. 102510.

[29] L. Kahl and U. G. Hofmann, "Comparison of algorithms to quantify muscle fatigue in upper limb muscles based on sEMG signals," *Med. Eng. Phys.*, vol. 38, no. 11, pp. 1260–1269, Nov. 2016.

[30] J. Hussain, K. Sundaraj, I. D. Subramaniam, and C. K. Lam, "Analysis of fatigue in the three heads of the triceps brachii during isometric contractions at various effort levels," *J. Musculoskeletal Neuronal. Interact.*, vol. 19, no. 3, pp. 276–285, Sep. 2019.

[31] K. Marri and R. Swaminathan, "Classification of muscle fatigue in dynamic contraction using surface electromyography signals and multifractal singularity spectral analysis," *J. Dyn. Syst., Meas., Control*, vol. 138, no. 11, Nov. 2016, Art. no. 111008.

[32] R. Fu, H. Wang, and W. Zhao, "Dynamic driver fatigue detection using hidden Markov model in real driving condition," *Expert Syst. Appl.*, vol. 63, pp. 397–411, Nov. 2016.

[33] F. N. Golabchi *et al.*, "Assessing aberrant muscle activity patterns via the analysis of surface EMG data collected during a functional evaluation," *BMC Musculoskeletal Disorders*, vol. 20, no. 1, p. 13, Jan. 2019.

[34] M. A. Oskoei and H. Hu, "Myoelectric control systems: A survey," *Biomed. Signal Process. Control, Elsevier*, vol. 2, no. 4, pp. 275–294, 2007.
[35] E. Scheme and K. Englehart, "Electromyogram pattern recognition for control of powered upper-limb prostheses: State of the art and challenges for clinical use," *J. Rehabil. Res. Develop.*, vol. 48, no. 6, p. 643, 2011.
[36] N. Mammone, C. Ieracitano, H. Adeli, A. Bramanti, and F. C. Morabito, "Permutation Jaccard distance-based hierarchical clustering to estimate EEG network density modifications in MCI subjects," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 29, no. 10, pp. 5122–5135, Oct. 2018.
[37] A. Gharehbaghi and M. Linden, "A deep machine learning method for classifying cyclic time series of biological signals using time-growing neural network," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 29, no. 9, pp. 4102–4115, Sep. 2018.
[38] A. Subasi and M. K. Kiymik, "Muscle fatigue detection in EMG using time–frequency methods, ICA and neural networks," *J. Med. Syst.*, vol. 34, no. 4, pp. 777–785, Aug. 2010.
[39] Y. Su, S. Sun, Y. Ozturk, and M. Tian, "Measurement of upper limb muscle fatigue using deep belief networks," *J. Mech. Med. Biol.*, vol. 16, no. 8, Dec. 2016, Art. no. 1640032.
[40] Z. Gao *et al.*, "EEG-based spatio–temporal convolutional neural network for driver fatigue evaluation," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 30, no. 9, pp. 2755–2763, Sep. 2019.
[41] H. Wu, Q. Huang, D. Wang, and L. Gao, "A CNN-SVM combined model for pattern recognition of knee motion using mechanomyography signals," *J. Electromyogr. Kinesiol.*, vol. 42, pp. 136–142, Oct. 2018.
[42] B. Baker, O. Gupta, N. Naik, and R. Raskar, "Designing neural network architectures using reinforcement learning," 2016, *arXiv:1611.02167*.
[43] U. C. Allard *et al.*, "A convolutional neural network for robotic arm guidance using semg based frequency-features," in *Proc. Intell. Robots Syst. (IROS)*, Daejeon, South Korea, Oct. 2016, pp. 2464–2470.
[44] V. Nair and G. E. Hinton, "Rectified linear units improve restricted Boltzmann machines Vinod Nair," in *Proc. Int. Conf. Int. Conf. Mach. Learn.*, 2010, pp. 1–8.
[45] C. J. C. H. Watkins, "Learning with delayed rewards," Ph.D. dissertation, Dept. Psychol., Univ. Cambridge, Cambridge, U.K., 1989.
[46] V. Mnih *et al.*, "Human-level control through deep reinforcement learning," *Nature*, vol. 518, pp. 529–533, Feb. 2015.
[47] Y. Chen, H. Jiang, C. Li, X. Jia, and P. Ghamisi, "Deep feature extraction and classification of hyperspectral images based on convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 10, pp. 6232–6251, Oct. 2016.
[48] C. J. C. Burges, "A tutorial on support vector machines for pattern recognition," *Data Mining Knowl. Discovery*, vol. 2, no. 2, pp. 121–167, 1998.
[49] N. E. Huang *et al.*, "The empirical mode decomposition and the Hilbert spectrum for nonlinear and non-stationary time series analysis," *Proc. Roy. Soc. London A, Math., Phys. Eng. Sci.*, vol. 454, no. 1971, pp. 903–995, Mar. 1998.
[50] Z. Zhang *et al.*, "A novel deep learning approach with data augmentation to classify motor imagery signals," *IEEE Access*, vol. 7, pp. 15945–15954, 2019.
[51] J. Dinarès-Ferran, R. Ortner, C. Guger, and J. Solé-Casals, "A new method to generate artificial frames using the empirical mode decomposition for an EEG-based motor imagery BCI," *Frontiers Neurosci.*, vol. 12, p. 308, May 2018.
[52] Z. Wu and N. E. Huang, "Ensemble empirical mode decomposition: A noise-assisted data analysis method," *Adv. Adapt. Data Anal.*, vol. 1, no. 1, pp. 1–41, 2008.
[53] M. E. Torres, M. A. Colominas, G. Schlotthauer, and P. Flandrin, "A complete ensemble empirical mode decomposition with adaptive noise," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, May 2011, pp. 4144–4147.
[54] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *Proc. 3th Int. Conf. Learn. Rep. (AISTATS)*, San Diego, CA, USA, 2015, pp. 1–15.
[55] X. Glorot, A. Bordes, and Y. Bengio, "Deep sparse rectifier neural networks," in *Proc. 14th Int. Conf. Art. Intel. Stat. (AISTATS)*, Fort Lauderdale, FL, USA, 2011, pp. 315–323.
[56] G. E. Hinton, N. Srivastava, A. Krizhevsky, I. Sutskever, and R. R. Salakhutdinov, "Improving neural networks by preventing co-adaptation of feature detectors," 2012, *arXiv:1207.0580*.
[57] J. R. Potvin and L. R. Bent, "A validation of techniques using surface EMG signals from dynamic contractions to quantify muscle fatigue during repetitive tasks," *J. Electromyogr. Kinesiol.*, vol. 7, no. 2, pp. 131–139, Jun. 1997.
[58] C. J. De Luca, L. D. Gilmore, M. Kuznetsov, and S. H. Roy, "Filtering the surface EMG signal: Movement artifact and baseline noise contamination," *J. Biomech.*, vol. 43, no. 8, pp. 1573–1579, 2010.
[59] M. R. Al-Mulla, F. Sepulveda, and M. Colley, "Evolved pseudo-wavelet function to optimally decompose sEMG for automated classification of localized muscle fatigue," *Med. Eng. Phys.*, vol. 33, no. 4, pp. 411–417, May 2011.
[60] G. Borg, "Psychophysical scaling with applications in physical work and the perception of exertion," *Scandin. J. Work, Environ. Health*, vol. 16, pp. 55–58, Jan. 1990.

Shurun Wang received the bachelor's degree in automation from the Hefei University of Technology, Hefei, China, in 2016, and the master's degree in control theory and control engineering from the School of Electrical Engineering and Automation, Hefei University of Technology, in 2019, where he is currently pursuing the Ph.D. degree with the School of Electrical Engineering and Automation.

His research interests include human–exoskeleton interaction, complex networks, and surface electromyography (sEMG) analysis.

Hao Tang (Member, IEEE) received the B.E. degree from the Anhui University of Science and Technology, Huainan, China, in 1995, the M.E. degree from the Institute of Plasma Physics, Chinese Academy of Sciences, Beijing, China, in 1998, and the Ph.D. degree from the University of Science and Technology of China (USTC), Hefei, China, in 2002.

He has been a Post-Doctoral Researcher with the Advanced Robotics with Artificial Intelligence Laboratory, The University of Tokyo, Tokyo, Japan, from 2005 to 2007. He is currently a Professor with the Hefei University of Technology, Hefei. His research interests include discrete event dynamic systems (DEDSs), reinforcement learning, the methodology of neurodynamic programming, and intelligent optimization.

Bin Wang received the bachelor's degree in automation from the Hefei University of Technology, Hefei, China, in 2015, and the master's degree in detection technology and automatic equipment from the School of Electrical Engineering and Automation, Hefei University of Technology, in 2018, where he is currently pursuing the Ph.D. degree with the School of Electrical Engineering and Automation.

His current research interests include discrete event dynamic systems and reinforcement learning.

Jia Mo received the bachelor's degree in automation from the Hefei University of Technology, Hefei, Anhui, China, in 2018, and the master's degree in control theory and control engineering from the School of Electrical Engineering and Automation, Hefei University of Technology, in 2021.

His current research interests include signal processing and machine learning.